

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

02

Q1

ALGEBRA

Which of the following ordered pairs (x, y) satisfies the inequality $5x - 3y < 4$?

1. $(1, 1)$
 2. $(2, 5)$
 3. $(3, 2)$
- A. I only
- B. II only
- C. I and II only
- D. I and III only

Q2

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$$x + y = 350$$

The given equation relates the total number of maple trees, x , and the total number of birch trees, y , planted in a 14-acre forest preserve. If 245 maple trees were planted in the forest preserve, how many birch trees were planted in the forest preserve?

- A. 14
- B. 25
- C. 105
- D. 245

Q3

ALGEBRA

If $\frac{1}{2}x - \frac{1}{6}x = 1$, what is the value of x ?

- A. -4
- B. $\frac{1}{3}$
- C. 3
- D. 6

x	$f(x)$
0	29
1	32
2	35

For the linear function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines $f(x)$?

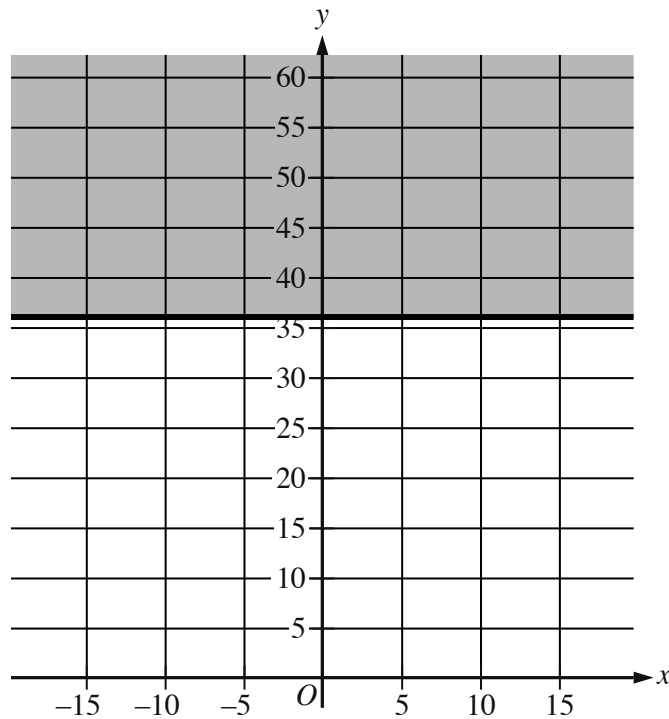
- A. $f(x) = 3x + 29$
- B. $f(x) = 29x + 32$
- C. $f(x) = 35x + 29$
- D. $f(x) = 32x + 35$

$$x = 5$$

$$y = x - 8$$

Which of the following points x, y is the solution to the given system of equations in the xy -plane?

- A. 0,0
- B. 5,-3
- C. 5,-8
- D. 5,8



- The boundary of the inequality is a solid line.
 - The line is horizontal.
 - The line passes through the following points:
 - (negative 15 comma 36)
 - (0 comma 36)
 - (15 comma 36)
- The area above the boundary is shaded.

The shaded region shown in the graph represents all the solutions to which inequality?

- A. $x \leq 36$
- B. $x \geq 36$
- C. $y \leq 36$

D. $y \geq 36$

Q7

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The equation $40x + 20y = 160$ represents the number of sweaters, x , and number of shirts, y , that Yesenia purchased for \$ 160. If Yesenia purchased 2 sweaters, how many shirts did she purchase?

- A. 3
- B. 4
- C. 8
- D. 40

Q8

ALGEBRA

If $8x = 6$, what is the value of $72x$?

- A. 3
- B. 15
- C. 54
- D. 57

$$P(t) = 250 + 10t$$

The population of snow leopards in a certain area can be modeled by the function P defined above, where $P(t)$ is the population t years after 1990. Of the following, which is the best interpretation of the equation $P(30) = 550$?

- A. The snow leopard population in this area is predicted to be 30 in the year 2020.
- B. The snow leopard population in this area is predicted to be 30 in the year 2030.
- C. The snow leopard population in this area is predicted to be 550 in the year 2020.
- D. The snow leopard population in this area is predicted to be 550 in the year 2030.

$$\begin{aligned}x + y &= 20 \\ 2(x + y) + 3y &= 85\end{aligned}$$

If (x, y) is the solution to the given system of equations, what is the value of y ?

- A. 10
- B. 15
- C. 60
- D. 65